

BSTA 511/611 textbook & software information

Textbook

The primary textbook for BSTA 511/611 will be *Introductory Statistics for the Life and Biomedical Sciences* by Vu and Harrington.

- The textbook website is <https://openintro.org/book/biostat/>, which includes links to acquire the book as either a pdf or paperback. The pdf of the textbook is free, other than an optional contribution.
- After acquiring the pdf version of the book, you will also have the option to download a tablet friendly version pdf with smaller margins – *I prefer the tablet version even on my computer.*

Software: R & RStudio

Many of the biostatistics courses in the SPH are taught using the statistical computing software R, and within the application RStudio. Below are links for downloading these programs, as well as resources to help get you started in learning how to use them. Both programs are open source and free.

Please make sure you have both R and RStudio installed for the first class meeting of BSTA 511, and work through the lessons listed in the “Get started with R and RStudio” section below.

1. Install R and RStudio
 - a. R and RStudio are two separate programs. You must first install R, and then RStudio Desktop. Even if you already have R installed, I recommend installing the latest version if it has been more than two years since you last updated it.
 - b. Install **R** (first)
 - i. Windows: Download from <https://cran.rstudio.com/bin/windows/base/> from the link “Download R 4.6.1 for Windows”
 - ii. Mac OS X: Download the latest .pkg file (top link, currently R-4.6.1.pkg) from <https://cran.rstudio.com/bin/macosx/>
 - c. Install **RStudio Desktop** (second)
 - i. Select the download file corresponding to your operating system from the RStudio IDE section of the webpage <https://docs.posit.co/ide/user/#rstudio-ide-oss-downloads>
 - d. Prefer tutorials or videos? Check out some of these options:
 - i. Kristen Sosulski’s *Installing R and RStudio (2024)*: <https://www.youtube.com/watch?v=3f8TMPOZTLE>
 - ii. OpenIntroOrg’s YouTube video: <https://www.youtube.com/playlist?list=PLkIselvEzpM73U-0ONpe7eHp2WCultVWQ>

2. Get started with R and RStudio

We will be using some of the class materials from the [Introduction to R class \(PUBH 523\)](#) for BSTA 511/611. The [schedule](#) for the Intro to R class includes links to recorded lesson videos, slides, and sometimes sections from textbooks (columns TB1 and TB2) for additional information.

For the first day of class, please work through the content from the following lessons:

- Lesson 1: Welcome
- Lesson 3: Download + Intro to R & RStudio
- Lesson 4: Basics
- Lesson 6: Quarto and .qmd’s